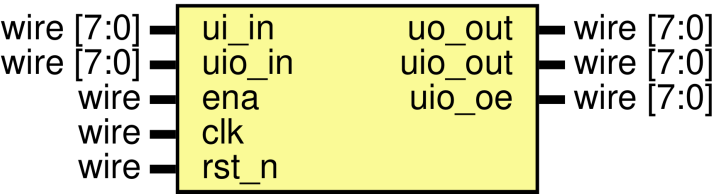


Entity: protocol_engine

- **File:** protocol_engine.v

Diagram



Ports

Port name	Direction	Type	Description
ui_in	input	wire [7:0]	Input-only protocol pins; <code>ui_in[3:0]</code> become logical pins 4-7.
uo_out	output	wire [7:0]	Protocol output latch, or host RX FIFO data during a host read.
uio_in	input	wire [7:0]	Host loader/FIFO controls on bits 0-3 and bidirectional protocol inputs on bits 4-7.
uio_out	output	wire [7:0]	Bidirectional protocol output values on bits 4-7; bits 0-3 are zero.

uio_oe	output	wire [7:0]	Per-pin output enables on bits 4-7; bits 0-3 are zero.
ena	input	wire	Harness enable input; retained for the standard wrapper interface and not used for control.
clk	input	wire	System clock used by the divider, loader, FIFOs, and execution FSM.
rst_n	input	wire	Active-low asynchronous reset.

Signals

Name	Type	Description
imem [0:31]	reg [15:0]	32-word instruction memory; reset fills it with HALT instructions.
cfg_clkdiv_int	reg [15:0]	Integer part of the system-clock divider.
cfg_clkdiv_frac	reg [7:0]	Fractional divider term accumulated to distribute extra cycles.
cfg_wrap_top	reg [4:0]	Highest program address included in the automatic wrap range.
cfg_wrap_bottom	reg [4:0]	Lowest program address included in the automatic wrap range.

cfg_shift_dir	reg [1:0]	Shift direction: zero emits LSB first; one emits MSB first.
cfg_autopull	reg	Enables TX FIFO refill while executing OUT.
cfg_autopush	reg	Enables RX FIFO capture while executing IN.
cfg_pull_thresh	reg [4:0]	Autopull threshold, tested when <code>osr_count</code> is at or below it.
cfg_push_thresh	reg [4:0]	Autopush threshold, tested when <code>isr_count</code> is at or above it.
cfg_side_count	reg [3:0]	Number of low side-set bits applied to protocol pins 0-3.
cfg_side_oe	reg	Selects side-set destination: output value when zero, output enable when one.
clkdiv_int_cnt	reg [15:0]	Current integer divider countdown.
clkdiv_frac_acc	reg [7:0]	Fractional divider accumulator.
tick	reg	One-cycle execution pulse generated when the divider expires.

clkdiv_frac_sum = {1'b0, clkdiv_frac_acc}	wire [8:0]	Fractional accumulator plus configured fraction, including carry.
pc	reg [4:0]	Program counter for the 32-word instruction memory.
instr	reg [15:0]	Instruction latched during S_FETCH and decoded during S_EXEC.
osr	reg [31:0]	Output shift register consumed by OUT and loaded by PULL/MOV.
isr	reg [31:0]	Input shift register filled by IN and whole-bus SAMPLE.
osr_count	reg [4:0]	Number of valid or remaining output bits used by autopull.
isr_count	reg [4:0]	Number of input bits accumulated since the last push.
x_reg	reg [7:0]	General-purpose scratch/count register used by jumps and delays.
y_reg	reg [7:0]	General-purpose scratch/count register used by jumps and delays.

pin_out	reg [7:0]	Latched logical protocol output values.
pin_oe	reg [7:0]	Latched logical protocol output enables; zero releases a pin.
delay_cnt	reg [15:0]	Countdown for S_DELAY and the wide DELAY instruction.
state	reg [1:0]	Current execution state: fetch, execute, or delay.
next_state	reg [1:0]	Next FSM state selected while executing an instruction.
irq_pending	reg	Sticky internal flag set by IRQ; no external IRQ port is exposed.
host_fifo_data	reg [7:0]	Latched RX FIFO byte returned on a host read.
pin_in_prev	reg [7:0]	Previous protocol input sample used for edge detection.
wait_rise_pending	reg [7:0]	Latched rising-edge events available to WAIT.
wait_fall_pending	reg [7:0]	Latched falling-edge events available to WAIT.

side_mask	reg [3:0]	Mask derived from <code>cfg_side_count</code> for same-cycle side-set updates.
tx_fifo [0:7]	reg [7:0]	Eight-entry host-to-engine transmit FIFO.
rx_fifo [0:7]	reg [7:0]	Eight-entry engine-to-host receive FIFO.
tx_wr	reg [3:0]	TX FIFO write pointer.
tx_rd	reg [3:0]	TX FIFO read pointer.
tx_count	reg [3:0]	Number of bytes currently in the TX FIFO.
rx_wr	reg [3:0]	RX FIFO write pointer.
rx_rd	reg [3:0]	RX FIFO read pointer.
rx_count	reg [3:0]	Number of bytes currently in the RX FIFO.
tx_empty = (tx_count == 4'd0)	wire	Indicates that no TX data is available for <code>PULL</code> or <code>autopull</code> .
rx_full = (rx_count == 4'd8)	wire	Indicates that the eight-entry RX FIFO cannot accept another byte.

host_fifo_mode = (~uio_in[3]) & uio_in[2]	wire	Selects the host FIFO access mode when the loader is inactive.
host_write_req = host_fifo_mode & uio_in[0]	wire	Host request to enqueue <code>ui_in</code> into the TX FIFO.
host_read_req = host_fifo_mode & uio_in[1]	wire	Host request to dequeue one byte from the RX FIFO.
rx_dequeue = host_read_req && (rx_count != 4'd0)	wire	Valid RX dequeue request, suppressed when the RX FIFO is empty.
protocol_in = {ui_in[3:0], uio_in[7:4]}	wire [7:0]	Logical protocol input bus; pins 0-3 come from <code>uio_in[7:4]</code> , pins 4-7 from <code>ui_in[3:0]</code> .
autopull_hit = cfg_autopull && (osr_count <= cfg_pull_thresh) && !tx_empty && (instr[15:12] == 4'h4)	wire	True when the current <code>OUT</code> can refill its OSR directly from TX FIFO.
autopush_hit = cfg_autopush && (isr_count >= cfg_push_thresh) && (!rx_full rx_dequeue) && (instr[15:12] == 4'h3)	wire	True when the current <code>IN</code> can commit its ISR byte to RX FIFO.
tx_fifo_word	wire [31:0]	TX FIFO byte positioned at the active end of the OSR for the shift direction.

osr_eff = autopull_hit ? tx_fifo_word : osr	wire [31:0]	OSR value visible to the current OUT, including same-cycle autopull data.
out_bit = cfg_shift_dir == 2'd0 ? osr_eff[0] : osr_eff[31]	wire	Bit emitted by OUT before the OSR shifts.
osr_count_eff = autopull_hit ? 5'd8 : osr_count	wire [4:0]	Effective OSR count, reset to eight when autopull supplies a byte.
load_shift	reg [15:0]	Serial shift register used while loading one instruction.
cfg_shift	reg [7:0]	Serial shift register used while loading one configuration byte.
load_bit	reg [3:0]	Number of instruction bits received in the current word.
cfg_bit	reg [2:0]	Number of configuration bits received in the current byte.
load_addr	reg [4:0]	Instruction or configuration address selected by the loader.
host_clk_d	reg	Delayed host clock used to detect loader clock rising edges.

load_mode	reg	Indicates that the engine is awaiting the falling edge of loader enable.
uio3_d	reg	Delayed loader-enable input used to detect the start of a load.
host_clk_rise = uio_in[1] & ~host_clk_d	wire	Rising edge of the host serial clock.
load_start = uio_in[3] & ~uio3_d	wire	Rising edge that starts a new instruction/configuration load at address zero.
cfg_byte = {cfg_shift[6:0], uio_in[0]}	wire [7:0]	Configuration byte assembled MSB first from the host data input.
opcode = instr[15:12]	wire [3:0]	Four-bit instruction operation code.
operand = instr[11:8]	wire [3:0]	Instruction-specific operand and condition fields.
side_set = instr[7:4]	wire [3:0]	Same-cycle side-set value, or control bits for WAIT and jump target bits.
delay_imm = instr[3:0]	wire [3:0]	Extra tick delay applied after the instruction.
jump_tgt = {operand[3], side_set}	wire [4:0]	Five-bit jump target assembled from operand and side-set fields.

wait_rise_event = protocol_in & ~pin_in_prev	wire [7:0]	Inputs that changed from low to high on the current system-clock sample.
wait_fall_event = ~protocol_in & pin_in_prev	wire [7:0]	Inputs that changed from high to low on the current system-clock sample.
wait_edge_cycle = !uio_in[3] && !load_mode && tick && state == S_EXEC && opcode == OP_WAIT	wire	Identifies a tick on which an edge-based WAIT may consume an event.
wait_rise_clear_mask = (wait_edge_cycle && side_set[0] && operand[0] && wait_rise_pending[operand[3:1]]) ? (8'b1 << operand[3:1]) : 8'b0	wire [7:0]	Rising-edge event bit consumed by the current WAIT, if any.
wait_fall_clear_mask = (wait_edge_cycle && side_set[0] && !operand[0] && wait_fall_pending[operand[3:1]]) ? (8'b1 << operand[3:1]) : 8'b0	wire [7:0]	Falling-edge event bit consumed by the current WAIT, if any.
tx_dequeue = tick && (state == S_EXEC) && ((opcode == OP_OUT && autopull_hit) (opcode == OP_PULL && !tx_empty))	wire	Indicates a TX FIFO byte consumed by PULL or autopull.
tx_enqueue = host_write_req && ((tx_count != 4'd8) tx_dequeue)	wire	Indicates a host TX enqueue, including same-cycle replacement of a consumed byte.
rx_enqueue = tick && (state == S_EXEC) && ((opcode == OP_PUSH && (!rx_full rx_dequeue)) autopush_hit)	wire	Indicates an RX FIFO byte produced by PUSH or autopush.

pc_target	reg [4:0]	Candidate next program counter before wrap-range correction.
next_pin_out	reg [7:0]	Temporary output-latch value combining the instruction and side-set effects.
i	integer	Loop variable used to initialize instruction memory on reset.
_unused = &{ena, 1'b0}	wire	Constant-low sink that prevents the unused enable input from being optimized as an undriven port.

Constants

Name	Type	Value	Description
S_FETCH		2'd0	Fetch the instruction at <code>pc</code> from instruction memory.
S_EXEC		2'd1	Decode and execute the latched instruction.
S_DELAY		2'd2	Count down <code>delay_cnt</code> before returning to fetch.
OP_NOP		4'h0	No operation; advance to the next instruction.
OP_JMP		4'h1	Unconditional or conditional jump using the operand variant.

OP_WAIT		4'h2	Stall until a selected input level or edge condition is satisfied.
OP_IN		4'h3	Shift one selected protocol input bit into the ISR.
OP_OUT		4'h4	Emit one OSR bit and shift the OSR; supports open-drain enable control.
OP_PUSH		4'h5	Copy the low ISR byte into the RX FIFO, stalling if it is full.
OP_PULL		4'h6	Load the OSR from the TX FIFO, stalling if it is empty.
OP_MOV		4'h7	Move data among X, Y, OSR, ISR, and the output latch.
OP_SET		4'h8	Write <code>side_set</code> to a selected register or pin group.
OP_IRQ		4'h9	Set the sticky internal interrupt-pending flag.
OP_DELAY		4'hA	Delay for the 16-bit count formed by { <code>x_reg</code> , <code>y_reg</code> }.
OP_TOGGLE		4'hB	XOR protocol output pins 0-3 with <code>side_set</code> .
OP_SAMPLE		4'hC	Capture all eight logical protocol inputs into the ISR.
OP_HALT		4'hF	Remain in execute state and stop advancing the program.

Processes

- `clock_divider`: (@(posedge clk or negedge rst_n))
 - **Type:** always
 - **Description:** Generates the integer-plus-fractional execution `tick` from the system clock.

- `side_mask_decode: ( @(*) )`
 - **Type:** always
 - **Description:** Converts the configured side-set width into a low-bit mask.
- `host_edge_capture: ( @(posedge clk or negedge rst_n) )`
 - **Type:** always
 - **Description:** Synchronizes the host clock and loader-enable inputs for edge detection.
- `loader_and_execution_fsm: ( @(posedge clk or negedge rst_n) )`
 - **Type:** always
 - **Description:** Implements instruction/configuration loading and the tick-driven protocol execution FSM.
- `host_fifo_transfer: ( @(posedge clk or negedge rst_n) )`
 - **Type:** always
 - **Description:** Performs host TX FIFO writes and RX FIFO reads, including the host-visible read-data latch.
- `fifo_occupancy: ( @(posedge clk or negedge rst_n) )`
 - **Type:** always
 - **Description:** Updates TX and RX FIFO occupancy counts for enqueue and dequeue events.

State machines

The execution FSM advances only on a divider-generated `tick`: `S_FETCH` latches an instruction, `S_EXEC` performs the operation or stalls on a wait/FIFO condition, and `S_DELAY` counts down explicit or per-instruction delay cycles. A completed instruction normally returns to `S_FETCH`, subject to wrap addressing and delay insertion.

state

S_DELAY

delay_cnt == 16'd0

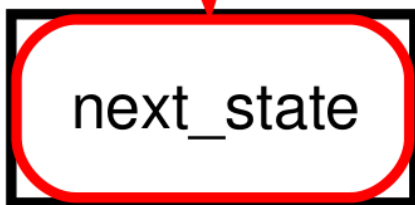
S_FETCH

S_EXEC

next_state != S_EXEC



next_state != S_EXEC



next_state